

CSE 251 – Electronic Circuits

Course Project: Design & Construction of a Mobile Phone Charger (5V DC Output)

Course Outcome: CO4

Program Outcome: PO1

Complex Engineering Problems: EP1, EP2 (see explanations at the end of this document)

Cognitive Level: C3

Psychomotor: P2, P3

Affective: A2

Knowledge Profile: K1, K3 (see explanations at the end of this document)

1. Project Overview

You are required to **design and build a compact mobile phone charger** using **low-level electronic components**, following real-world engineering constraints. Your design must convert 220 V AC (input) to a stable **5 V DC USB output** (USB-B/USB-C). The final system must be **compact, safe, and aesthetically packaged** with **no loose wires**. All components must be assembled neatly, safely, and compactly within limited space.

This project is assessed under Complex Engineering Problem level EP1 & EP2, Cognitive (C3), Psychomotor (P2, P3), and Affective (A2) skills.

2. Required Components

Your charger may use the following components:

- AC Plug + step-down transformer
- Bridge rectifier ($4 \times$ diodes or IC)
- Filter capacitor
- Voltage regulator IC or Zener diode
- USB-B or USB-C output connector
- Supporting resistors, wiring, breadboard and packaging materials
- Compact and aesthetically pleasing packaging

3. Technical Requirements

3.1 Electrical Design Requirements

1. Step-down stage: Select an appropriate transformer rating based on required DC output power.
2. Rectification stage: Use a full bridge rectifier.
3. Filtering stage: Choose filter capacitor(s) to minimize ripple.
4. Regulation stage: Use a Zener-diode regulator or linear IC regulator (e.g., 7805).
5. Output port: Provide standard USB-B or USB-C connector.
6. Wattage calculation (Mandatory):
 - Compute expected wattage = $V_{out} \times I_{out}$

- Verify wattage experimentally.
- Compare theoretical, simulated, and measured values.

3.2 Mechanical & Packaging Requirements

1. All wires must be solidly soldered, tightly fixed, *no loose/dangling components*.
2. Circuit must be placed inside a compact enclosure.
3. Final device must look clean, neat, and aesthetically packaged.
4. Clear labeling (AC IN / DC OUT) is encouraged.

4. Complex Engineering Problem (CEP) Requirements

EP1 – Depth of Knowledge Required:

Show that the problem **cannot be solved without applying in-depth electronics knowledge**.

1. Calculate the **maximum wattage** of the charger (input power, output power & efficiency).
2. Perform first-principles analytical design by showing calculations related to:
 - Transformer turns ratio selection
 - Peak rectifier output voltage
 - Ripple voltage estimation
 - Capacitor value calculation
3. Justify component values using **K3-level engineering knowledge** (physics of diodes, regulators, filters, etc.). Provide deeper justification for design choices.
4. Simulate the design (Multisim / Proteus / Falstad / LT-Spice) using the calculated values and compare.
5. State **how EP1 criteria were achieved**.

EP2 – Conflicting Requirements:

Compare at least two conflicting issues (EP2 requirement).

1. Provide **design proposals** showing how you resolved or optimized the tradeoffs (e.g., selecting a regulator over a Zener due to heat/power tradeoffs).
2. Demonstrate the conflicts you had to manage, such as:
 - High current vs compact size
 - Heat dissipation vs small space
 - Cost vs performance
 - Ripple reduction vs component size
 - Safety vs simplicity

Requirement	Conflict
High current output	Higher heat dissipation in regulator
Large filter capacitor (low ripple)	Limited physical space in enclosure

Compact size	Limited airflow, increased temperature
Low cost	Component quality constraints

5. Safety Notes

- Handle AC input carefully.
- Do NOT touch exposed AC terminals.
- Use heat-shrink tubes and insulation.
- Use a fuse if possible.

6. Grading Rubric (Total = 10 Marks)

Cognitive (C3): Applying – 4 Marks (From report)

- Correct design calculations
- Wattage computation & verification
- Justified selection of components
- EP1: Uses ≥ 2 levels of engineering knowledge with acceptable elaboration

Psychomotor (P2 + P3): Construction Quality – 4 Marks (From Demonstration and viva)

- Working solution
- Tight wiring (no loose wires)
- Compact & aesthetic packaging
- Accurate, working hardware output

Affective (A2): Responding – 2 Marks (From Demonstration and viva)

- Professionalism
- Safety practices
- Clear report writing
- Timely submission

EP1 + EP2 Achievement (Included Across Marks)

Your score will reflect how well you:

- EP1: Analyze problem using K3-level knowledge & evaluate multiple constraints
- EP2: Compare conflicting engineering requirements & provide 1–2 justified proposals

7. Project Report Requirements

Your report must follow the CSE251 project template. It should contain:

1. A cover page with group members details
2. Project Overview
 - What you are designing (a compact phone charger).

- Why proper rectification, filtering, and regulation is needed.

3. Design Details

- Calculate the **maximum wattage** of the charger (input power, output power & efficiency).
- Perform first-principles analytical design by showing calculations related to:
 - Transformer turns ratio selection
 - Peak rectifier output voltage
 - Ripple voltage estimation
 - Capacitor value calculation
- Justify component values using **K3-level engineering knowledge** (physics of diodes, regulators, filters, etc.). Provide deeper justification for design choices.

4. Circuit Diagram

- Must be drawn using any software of your choice..

5. Simulation Results

- Simulate rectifier + filter + regulator using any software of your choice.
- Include screenshots.

6. Hardware Construction

- Show photographs of soldering, packaging, and final compact design.

7. Measurement & Verification

- Output voltage with load.
- Measured wattage.
- Compare theoretical, simulation, and measured values.

8. EP1–EP2 Section (Mandatory)

- One dedicated section explaining how your project meets EP1 and EP2.

Explanations of Knowledge Profiles (K1, K3) and Complex Engineering Problems (EP1, EP2):

Range of Complex Engineering Problem Solving

Complex engineering problem solving as required in PO1, PO2, PO3, PO4, PO5, PO6, and PO7 are problems that have EP1 and some or all of EP2 to EP7 characteristics.

Attribute	Characteristics of Complex Engineering Problems
EP1: Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6, or K8 which allows a fundamental-based, first principles analytical approach
EP2: Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering and other issues
EP3: Depth of analysis required	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models
EP4: Familiarity of issues	Involve infrequently encountered issues
EP5: Extent of applicable codes	Are outside problems encompassed by standards and codes of practice for professional engineering
EP 6: Extent of stakeholder involvement and conflicting requirements	Involve diverse groups of stakeholders with widely varying needs
EP7: Interdependence	Are high level problems including many component parts or sub-problems

Knowledge Profile	Attribute
K1: Theory-based natural sciences	A systematic, theory-based understanding of the natural sciences applicable to the discipline
K2: Conceptually-based mathematics, numerical analysis, statistics, and formal aspects of computer and information science	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline
K3: Theory-based engineering fundamentals	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
K4: Forefront engineering specialist knowledge for practice	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
K5: Engineering design	Knowledge that supports engineering design in a practice area
K6: Engineering practice (technology)	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
K7: Comprehension of engineering in society	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability
K8: Research literature	Engagement with selected knowledge in the research literature of the discipline